

Full Name:	Aiyshwarya Aruchamy
Amex Contractor ID:	C7336090
AMEX Designation (as per AMEX role qualification):	Amex Role – Engineer II
Education qualification: highest degree completion - along with specialization	Integrated Master's Degree in Data Science
Year of completion for the highest degree with month:	Highest Degree completion with year and month
University name:	Amrita Vishwa Vidyapeetham
Total work experience (Overall experience - in months):	14 months (Aug 2023 – Present)
Skill set:	Technical skillset • Strong experience in Python for data engineering, machine learning, and backend development. • Hands-on experience with Java for object-oriented programming and application development. • Proficient in SQL including MySQL and PostgreSQL for data querying, optimization, and analysis. • Experience working in Linux-based environments for development and deployment. • Hands-on experience in Machine Learning algorithms including supervised and unsupervised learning. • Strong knowledge of Deep Learning architectures such as CNNs, Transformers, and GANs. • Experience working with Large Language Models (LLMs) and prompt engineering techniques. • Practical exposure to Natural Language Processing (NLP) tasks including sentiment analysis and OCR. • Experience with computer vision pipelines for image classification, detection, and segmentation. • Proficient with frameworks such as PyTorch, TensorFlow, Keras, and Scikit-

learn.

- Experience using Hugging Face and LangChain for AI and LLM-based applications.
- Hands-on experience with OpenCV for image preprocessing and augmentation.
- Strong data handling experience using Pandas, NumPy, and statistical modeling techniques.
- Experience in data visualization using Tableau, Power BI, Matplotlib, and Seaborn.
- Exposure to cloud platforms including AWS (S3, EC2), Azure, and Google Cloud.
- Experience in containerization using Docker for scalable deployments.
- Exposure to Kubernetes for orchestration and production deployment.
- Familiarity with big data tools such as Spark and Hadoop.
- Experience with Git for version control and collaborative development.

Soft skillset

- Strong analytical and problem-solving skills.
- Ability to work independently with minimal supervision.
- Excellent verbal and written communication skills.
- Strong collaboration skills within cross-functional teams.
- Experience mentoring junior team members.
- Ability to understand business requirements and translate them into technical solution.
- Strong attention to detail and quality standards.
- Proven ability to work in Agile and fast-paced environments.
- Experience handling multiple tasks and priorities simultaneously.
- Strong ownership mindset for deliverables

	and timelines. • Ability to adapt quickly to new technologies and tools. • Demonstrated leadership skills while handling team responsibilities. • Strong documentation and reporting skills. • Effective stakeholder communication and coordination. • Strong time management and organizational skills. • Ability to work under tight deadlines and pressure. • Proactive approach towards continuous learning and improvement. • Experience participating in sprint planning and retrospectives. • Ability to troubleshoot and debug complex technical issues. • High level of professionalism and compliance awareness.
Professional experiences	• Designed and developed modular deep learning pipelines for real-world computer vision use cases. • Built YOLO-based object detection and classification systems achieving high accuracy on production datasets. • Optimized deep learning models for edge deployment under constrained CPU/GPU environments. • Reduced inference latency by optimizing model architecture and deployment strategies. • Improved model robustness through advanced data augmentation and preprocessing techniques. • Worked extensively on image classification, object detection, and segmentation models. • Implemented Faster R-CNN, FCOS, ResNet, and Vision Transformer-based solutions. • Optimized inference using CUDA and TensorRT for production workloads. • Improved OCR pipelines for camera-based

	streams in industrial environments. • Collaborated closely with cross-functional teams including data engineers and product stakeholders. • Led a team of engineers for feature development and delivery. • Ensured adherence to project timelines and quality standards. • Deployed AI solutions on AWS infrastructure for scalable processing. • Implemented Docker-based containerization for reproducible environments. • Worked on real-time video analytics and streaming pipelines. • Developed Retrieval-Augmented Generation (RAG) systems using LangChain and Hugging Face. • Implemented sentiment analysis pipelines using NLP and LLM-based techniques. • Designed synthetic data generation pipelines for dataset augmentation. • Built automated annotation transfer tools reducing manual effort significantly. • Performed statistical analysis and validation of model performance. • Maintained version control and documentation using Git. • Ensured compliance with coding standards and security guidelines. • Participated in Agile ceremonies including sprint planning and reviews.
Certification (PMI/Scrum Master/testing certification etc.)	Currently no formal certification
Employment details: (Capture all past Employment details as per the overall work experience along with company name, start date, end date and role/ Designation)	Company name 1: Altair Engineering India Private Limited Start date: 01-Aug-2023 End date: 31-Jul-2024 Role/Designation: Data Analysis / AIoT Support Intern Company name 2 : Wastefull Insights Start date: 01-Nov-2024 End date: 01-Nov-2025 Role/Designation: Computer Vision Engineer

	Company name 3: Infosys Limited Start date: 12-11-2025 End date: Till date Role/Designation: Engineer II
Gap details: Mention if there are any Gap details from EDUCATION-EMPLOYMENT or in between EMPLOYMENT Experience details	No gap between education and employment No gap between employment periods